Can the flow regimes be told apart from the coarse observations alone?

Short study, 12 August 2026. Uses only the observed 64-by-64 fields - exactly the information a deployed model receives. Nine regimes, Reynolds 1000 to 8000, 180 sequences in total. The question behind it: is there enough regime information in a coarse input to condition a single reconstruction model on?

1. To the eye: no

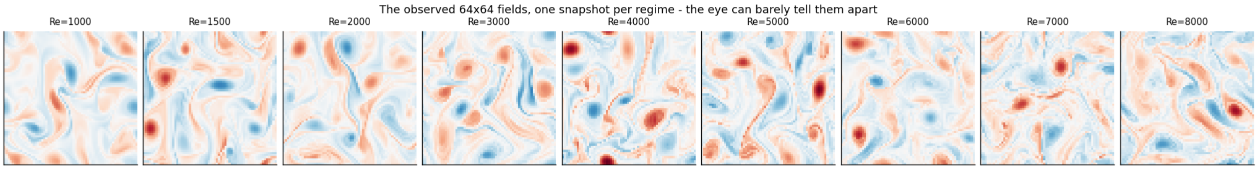

Figure 1. One observed snapshot per regime. The large scales are set by the forcing, which is identical everywhere; visually the regimes are nearly indistinguishable.

2. To the spectrum: yes, smoothly

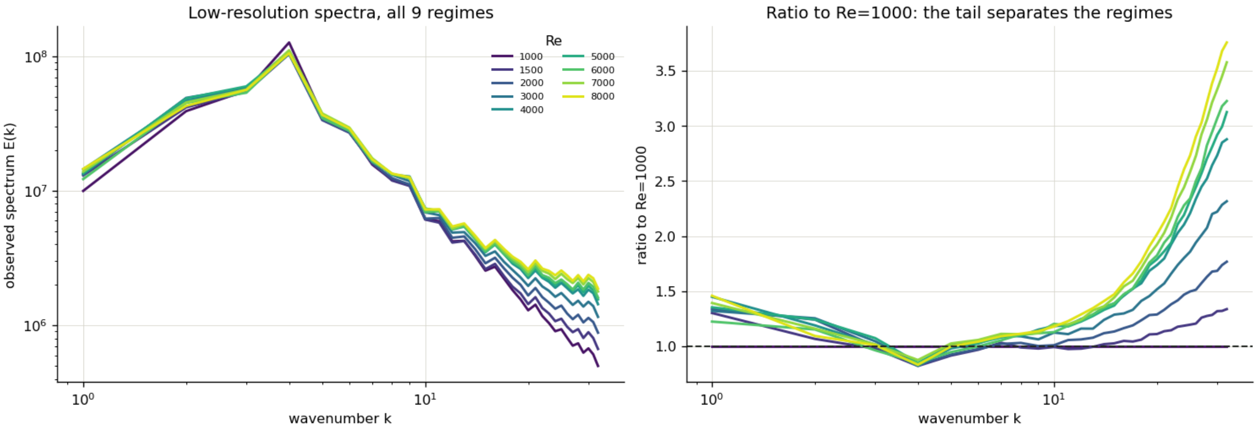

Figure 2. Left: the observed spectra of all nine regimes. Right: the same divided by the Reynolds 1000 curve. Below wavenumber ten the curves are almost identical; approaching the observation limit they fan out monotonically, reaching three point seven times the Reynolds 1000 energy at Reynolds 8000. The regime signature lives entirely in the observable tail.

3. Quantified: ordered, not categorical

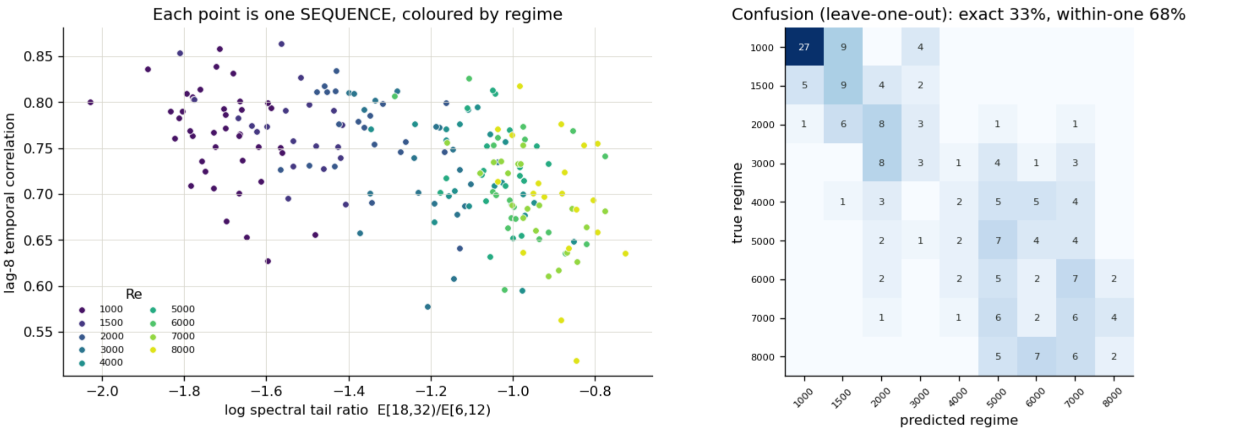

Figure 3. Left: each point is one sequence, described by two coarse-data observables. Right: leave-one-sequence-out classification over the nine regimes.

exact regime identification	33 percent (chance: 11 percent)
within one neighbouring regime	68 percent
rank correlation of predicted vs true regime	+0.82

The confusion matrix is band-diagonal: errors are almost always neighbouring regimes, and above Reynolds 5000 the regimes blur together - the dissipation range has moved past the observation limit, so the coarse data genuinely carries less and less distinction up there. Reynolds 1000 through 4000 separate well.

4. What this means for conditioning

Exact regime identity is not recoverable from a coarse observation - but conditioning does not need it. What the data supports is a smooth, monotone, continuous regime variable: the spectral tail ratio (or the closely related base-deficit statistic already used to choose training temperature) orders the regimes at rank correlation 0.82 from a single sequence, and averaging over even a few inputs sharpens it further. A model conditioned on such a variable receives essentially everything the observation knows about the regime. Two consequences. First, a conditioned model cannot be expected to distinguish Reynolds 6000 from 7000 - the information is not in its input - but the reconstruction difference between those regimes is correspondingly small, so the error committed is bounded by the same blurring. Second, the signal is already strong enough that an implicitly conditioned model - one simply trained across all regimes on raw inputs, letting the network read the tail itself - is worth testing before adding any explicit conditioning channel. That experiment (a single fine-tune with regime-rotating batches, each regime rewarded against its own spectral anchor) is prepared and queued behind the run currently on the accelerator.